

Yuyi Yang

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EDUCATION

Washington University in St. Louis | MO, USA

2023 – Present

- **Ph.D. of Computational and Data Science** (Social Work and Public Health Concentration)
- **Scholarship/Honors:** Dean's Select PhD Fellowship; 1st Place 2026 American Academy of Health Behavior 3 Minute Thesis Competition; McDonnell International Scholars Academy

Washington University in St. Louis | MO, USA

2020 – 2023

- **Master of Public Health** in Epidemiology/Biostatistics
- **Scholarship/Honors:** Brown School Scholarship

Fudan University | Shanghai, China

2015 – 2020

- **Bachelor of Medicine** in Basic Medical Sciences
- **Scholarship/Honors:** Undergraduate Academic Excellence Scholarship (2016, 2017); Fudan University Medical School Scholarship sponsored by Eisai (2016); UBC full scholarship visiting scholar sponsored by Shanghai government (SXF820019) and Fudan University Medical School (IBF101101) (2018).

RESEARCH INTERESTS

My research develops agentic AI systems and large language models for simulation and decision making in public health, health economics, and social systems. I build multi-agent workflows that integrate LLMs, retrieval-augmented generation, and machine learning to model individual behavior, collective choice, and information diffusion in networks, supporting scalable and theory-informed analysis of economic preferences, policy response, and behavioral change. I also study responsible AI for mental health services and methods to improve the reliability, interpretability, and cultural sensitivity of AI-assisted healthcare decision making.

Primary Advisor: Dr. Ruopeng An (<https://socialwork.nyu.edu/faculty-and-research/our-faculty/ruopeng-an.html>)

Secondary Advisor: Dr. Jiaxin Huang (<https://engineering.washu.edu/faculty/Jiaxin-Huang.html>)

PUBLICATION LISTS

Refereed Journal Publications

1. **Yang Y***, Huang J, Yang F, Cao Z, Fang Y, Ji M, Yang Y, Huang L, An R. (in press). A Scalable Multi-Agent Large Language Model Workflow to Support Grounded Theory Coding. *Journal of Social Computing*. 2026.
2. Lindsay R, **Yang Y**, An R. (in press). A Pilot Study of Empathy Helper, a Generative Artificial Intelligence Platform for Communication Practice in Graduate Social Work. *Journal of Social Work Education*. 2026.
3. Rfat M, **Yang Y** (co-first author), An R. Harnessing AI to Enhance Inclusion for Refugees with Disabilities: Mitigating Bias, Promoting Equity in Humanitarian and Resettlement Efforts. *Disability and the Global South*. 2026. 13(1), v1-v14. <https://dgsjournal.org/wp-content/uploads/2026/04/voices-2.pdf>
4. Li H, Huang J, Ji M, **Yang Y**, An R. Use of Retrieval-augmented Large Language Model for COVID-19 Fact-checking. *Journal of Medical Internet Research*. 2025. 27, e66098. <https://doi.org/10.2196/66098>
5. **Yang Y***, Wang C, Xiang X, An R. AI Applications to Reduce Loneliness Among Older Adults: A Systematic Review of Effectiveness and Technologies. *Healthcare*. 2025. 13(5), 446. <https://doi.org/10.3390/healthcare13050446>
6. **Yang Y**, Alba C, Wang C, Wang X, Anderson J, An R. GPT Models Can Perform Thematic Analysis in Public Health Studies, Akin to Qualitative Researchers. *Journal of Social Computing*, 2024. 5(4), 293-312. <https://ieeexplore.ieee.org/abstract/document/10832545>
7. Si Y, **Yang Y**, Wang X, Zu J, Chen X, Fan X, An R, Gong S. Quality and Accountability of ChatGPT in Health Care in Low-and Middle-Income Countries: Simulated Patient Study. *Journal of Medical Internet Research*. 2024. 26, e56121. <https://doi.org/10.2196/56121>
8. An R, Perez-Cruet J, Wang X, **Yang Y*** (corresponding author). Build Deep Neural Network Models to Detect Common Edible Nuts from Photos and Estimate Nutrient Portfolio. *Nutrients*. 2024. 16(9), 1294. <https://doi.org/10.3390/nu16091294>
9. **Yang Y***, Lin N, Batcheller Q, Zhou Q, Anderson J, An R. Sentiment Analysis of Tweets on Menu Labeling Regulations in the US. *Nutrients*. 2023. 15(19), 4269. <https://doi.org/10.3390/nu15194269>

10. An R, Shen J, Wang J, **Yang Y**. A Scoping Review of Methodologies for Applying Artificial Intelligence to Physical Activity Interventions. *Journal of Sport and Health Science*. 2023. <https://doi.org/10.1016/j.jshs.2023.09.010>
11. An R, **Yang Y*** (corresponding author), Yang F, Wang S. Use Prompt to Differentiate Text Generated by ChatGPT and Humans. *Machine Learning with Applications*. 2023. <https://doi.org/10.1016/j.mlwa.2023.100497>
12. An R, Batcheller Q, Wang J, **Yang Y**. Build Neural Network Models to Identify and Correct News Headlines Exaggerating Obesity-related Scientific Findings. *Journal of Data and Information Science*. 2023. <https://doi.org/10.2478/jdis-2023-0014>
13. An R, **Yang Y** (co-first author), Batcheller Q, Zhou Q. Sentiment Analysis of Tweets on Soda Taxes. *Journal of public health management and practice*. 2023. 29(5), 633–639. <https://doi.org/10.1097/PHH.0000000000001721>
14. **Yang Y**, Liu X. [Research process of Gasdermin E in inducing cell pyroptosis]. *Chinese Journal of Pathology*. 2021. 50(4), 421–424. <https://doi.org/10.3760/cma.j.cn112151-20200724-0058>

Refereed Computer Science Conference Publications

15. **Yang Y**, Huang L, Li J, Huang C, An Z, Xie Y, An R, Huang J. (2026). Contingent Valuation as a Benchmark for LLM Economic Decision-Making. Submitted to Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP).
16. An Z, Xie Y, Yu X, **Yang Y**, Chowdhury A, Qiu P. (2026). LUMINA: Longitudinal Multimodal Reasoning Agent for Medical Imaging. Submitted to Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP).
17. Huang L, Huang C, Li J, Cai D, **Yang Y**, Huang J. (2026). Nonsense Helps: Prompt Space Perturbation Broadens Reasoning Exploration. Submitted to Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP). Preprint available at: *arXiv:2605.05566*. <https://arxiv.org/abs/2605.05566>
18. Li J, Huang L, Huang C, Xu S, Cai D, **Yang Y**, Zhang M, Huang J. (2026). Process Rewards with Learned Reliability. Submitted to Proceedings of the Conference on Neural Information Processing Systems (NeurIPS). Preprint available at: *arXiv:2605.15529*. <https://arxiv.org/abs/2605.15529>

Invited Book Chapters

19. **Yang Y***, Traube D. (in press). AI in Mental Health Services. In: An R, Lindsey M. (Eds.), *Artificial Intelligence in Social Work: Bridging Technology and Humanity*. Springer.
20. **Yang Y***, Huang J, An R. (in press). AI in Social Work Research. In: An R, Lindsey M. (Eds.), *Artificial Intelligence in Social Work: Bridging Technology and Humanity*. Springer.
21. **Yang Y***, An R, Lindsey M. (in press). Conclusion—Challenges and Opportunities for AI in Social Work. In: An R, Lindsey M. (Eds.), *Artificial Intelligence in Social Work: Bridging Technology and Humanity*. Springer. <https://doi.org/10.1007/978-3-032-18443-6>
22. **Yang Y***, An R. (2025). Dementia Care and Generative Artificial Intelligence. In: Bennett G, Goodall E. (Eds.), *The Palgrave Encyclopedia of Disability*. Palgrave Macmillan, Cham. https://doi.org/10.1007/978-3-031-40858-8_528-1

Under Review/Revision

23. Ji M, Yang F, **Yang Y**, Tebeje H, Ssewamala F, An R. Comparing Large Language Models and Machine Learning for Diabetes Prediction in Data-Limited Public Health Contexts. (under review)
24. Zhu Y, **Yang Y**, Thompson R. Using Transformer-Based Models to Monitor and Understand Momentary Affect Intensity from Smartphone Sensors. *JMIR mHealth and uHealth*. (under revision)
25. Huang J, **Yang Y**, Ji M, An R. Large Language Model-Based Prediction of Severe Outcomes in VAERS Reports for Vaccine Safety Surveillance. *PLOS Digital Health*. (under revision)
26. **Yang Y***, Yang F, An R, Luke D. Countering HPV-Vaccine Misinformation: An Integrative Cognitive-Affective Framework for Theory-Driven Digital Health Education Interventions. *Health Education & Behavior*. (under revision)
27. Huang J, **Yang Y**, Ji M, Alba C, Zhang S, An R. (2025). Use of Retrieval-Augmented Large Language Model Agent for Long-Form COVID-19 Fact-Checking. *arXiv preprint arXiv:2512.00007*. <https://arxiv.org/abs/2512.00007>
28. Tey SL, An R, **Yang Y**, Huynh DTT, Baggs G, Deutz NEP, Cruz-Jentoft AJ, Atherton PJ, Wang X, Chew STH, Pereira SL. Effects of Beta-Hydroxy-Beta-Methylbutyrate (HMB) Interventions on Muscle Health in Community-Dwelling Adults Aged 55+ Years: A Systematic Review and Meta-Analysis. (under review)

REFEREED ABSTRACTS & PRESENTATIONS AT CONFERENCES

1. **Yang Y**, Huang L, Huang J, An R. Persona-Based LLM Agents for Replicating U.S. Willingness-to-Pay per QALY Estimates from a Published Survey. 2026 Annual Meeting and Expo by the American Public Health Association (APHA), San Antonio, Texas

2. Huang J, **Yang Y**, Ji M, An R. Predicting Severe Vaccine Adverse Events Using Large Language Models with Self-Refinement. 2026 Annual Meeting and Expo by the American Public Health Association (APHA), San Antonio, Texas
3. Tey S, An R, **Yang Y**, Huynh D, Baggs M, Deutz N, Cruz-Jentoft A, Atherton P, Wang X, Chew S, Pereira S. Beta-Hydroxy-Beta-Methylbutyrate (HMB) Interventions Improved Muscle Mass and Strength in Community-Dwelling Adults Aged 55+: A Systematic Review and Meta-analysis of Randomized Controlled Trials. The 48th European Society for Clinical Nutrition and Metabolism (ESPEN 2026) Congress, Berlin, Germany
4. Lai PHL, **Yang Y**. Learning with AI in Later Life: Educational Inequality in ChatGPT Use Among U.S. Adults Aged 50 and Older. International Conference on Gerontechnology (ICG) 2026, Hong Kong, China
5. **Yang Y**, Huang J, Yang F, Cao Z, Fang Y, Ji M, Yang Y, Huang L, An R. A Scalable Multi-Agent Large Language Model Workflow for Grounded Theory Coding. 2026 Scientific Meeting by the American Academy of Health Behavior (AAHB), Austin, Texas
6. Huang J, **Yang Y**, Ji M, Alba C, Zhang S, An R. Use of Retrieval-Augmented Large Language Model Agent for Long-Form COVID-19 Fact-Checking. 2026 Scientific Meeting by the American Academy of Health Behavior (AAHB), Austin, Texas
7. Zhu Y, **Yang Y**, Thompson R. Using Transformer-Based Models to Monitor and Understand Momentary Affect Intensity from Smartphone Sensors. 2026 Society for Affective Science (SAS) Annual Conference, Pittsburgh, Pennsylvania
8. Tey S, An R, **Yang Y**, Pereira S, Dieu T, Baggs M. Effects of β -hydroxy- β -methylbutyrate (HMB) Interventions on Muscle Health Outcomes in Community-dwelling Adults Aged 55 Years and above: A Systematic Review and Meta-analysis. The International Conference on Frailty and Sarcopenia Research (ICFSR) 2026, Washington D.C.
9. **Yang Y**, Alba C, An R. Empowering Social Work Research with GPT: A Framework for Scalable, Ethical Thematic Analysis in Public Health Studies. Society for Social Work and Research (SSWR) 30th Annual Conference (2026), Washington D.C.
10. **Yang Y**, Yang F, Luke D, An R. Countering HPV-Vaccine Misinformation: An Integrative Cognitive-Affective Framework for Theory-Driven Digital Interventions. 2025 National Conference on Health Communication, Marketing and Media (NCHCMM), Atlanta, Georgia
11. Zhu Y, **Yang Y**, Thompson R. Predicting Momentary Affect and Emotion Differentiation with Passive Sensors: The Benefit of Incorporating Self-Reported Location Information into GPS Data. The Society for Research in Psychopathology (SRP) 2024 Annual Meeting, Montreal, Canada
12. An R, Perez-Cruet J, Wang X, **Yang Y**. Build Deep Neural Network Models to Detect Common Edible Nuts from Photos and Estimate Nutrient Portfolio. 2024 Institute of Clinical and Translational Sciences (ICTS) Data Research Symposium, St. Louis, Missouri
13. **Yang Y**, Lin N, Batcheller Q, Zhou Q, Anderson J, An R. Sentiment Analysis of Tweets on Menu Labeling Regulations in the US. 2024 Scientific Meeting by the American Academy of Health Behavior (AAHB), Savannah, Georgia
14. An R, **Yang Y** (co-first author), Batcheller Q, Zhou Q. Sentiment Analysis of Tweets on Soda Taxes. 2023 Institute of Clinical and Translational Sciences (ICTS) Data Research Symposium, St. Louis, Missouri
15. An R, Batcheller Q, Wang J, **Yang Y**. Build Neural Network Models to Identify and Correct News Headlines Exaggerating Obesity-related Scientific Findings. 2023 Institute of Clinical and Translational Sciences (ICTS) Data Research Symposium, St. Louis, Missouri

OTHER PRESENTATIONS & GUEST LECTURES

Competitions

1. 1st Place, 2026 American Academy of Health Behavior (AAHB) 3 Minute Thesis Competition, Austin, Texas
2. Top 8 Finalist, 2024 Transdisciplinary Institute in Applied Data Sciences (TRIADS) Lightning Talk Competition, St. Louis, MO

Guest Lectures

1. Discussion on AI and Generative AI in Public Health Research. 2026. International Center for Child Health and Development (ICHAD) Summer Intensive Training Session, Washington University in St. Louis, MO
2. Applying Artificial Intelligence to Policy Analysis. 2025. Invited by Dr. Domingo Morel, Wagner Graduate School of Public Service, New York University, NY

Presentations at Conference

1. Zhu Y, **Yang Y**, Thompson R. Using Transformer-based Models to Monitor and Understand Momentary Affect Intensity from Smartphone Sensors. 2025 WashU Mobile Sensing Conference, St. Louis, MO

2. **Yang Y**, Alba C, An R. GPT Models Can Perform Thematic Analysis in Public Health Studies, Akin to Qualitative Researchers. 2024 AI for Health Symposium, St. Louis, MO
3. An R, **Yang Y**, Yang F, Wang S Use Prompt to Differentiate Text Generated by ChatGPT and Humans. 29th WashU Annual Graduate Research Symposium (2024), St. Louis, MO
4. **Yang Y**, Lin N, Batcheller Q, Zhou Q, Anderson J, An R. Sentiment Analysis of Tweets on Menu Labeling Regulations in the US. 2023 The Public Health at WashU Annual Conference, St. Louis, MO

WORK EXPERIENCES

O'Reilly Auto Parts Louis | MO, USA

2025 Summer

AI/ML PhD Intern

- Optimized a multimodal visual search system on GCP Vertex AI using CLIP-based image-text embeddings, boosting Top-K accuracy by ~20%.
- Designed a semantic similarity evaluation pipeline achieving 0.90 F1 / 0.96 AUC on internal benchmark datasets.

Washington University in St. Louis | MO, USA

Instructor, Mentored Teaching Experience (MTE), Teaching Assistant (TA)

- S55/S90.MPH.5139 Applied Machine Learning Using Health Data (Instructor). 2025 Spring
- S55/S90.MPH.5601 Applied Deep Learning Using Health Data (Instructor). 2024 Fall
- 2023 Artificial Intelligence Applications for Health Data Advanced Learning Certificate (MTE). 2023 Fall
- Applied Machine Learning Using Health Data (TA). 2023 Spring

DasCapital Overseas Limited | Shanghai, China

2020 – 2022

Full-time Associate, Full-time Analyst, Intern

- Financial advisory for private placements of growth-stage medical enterprises; produced 100+ pitchbooks, 10 business plans, and 8 industry reports; supported 2021 transactions totaling \$200M+.
- Selected transactions: Nutshell Therapeutics (Series A/A+), CommBio Therapeutics (Corporate Round/Pre-A), Sciwind Biosciences (Series C), Holosensor Medical (Pre-A+), Forqaly Medical (Series B).

Jingan District Central Hospital of Shanghai | Shanghai, China

2018 – 2019

Undergraduate Intern

- Rotated through Obstetrics and Gynecology, Internal Medicine, Surgery, and Pediatrics; supported clinicians in inpatient and outpatient settings.
- Assisted in surgical procedures and perioperative coordination, including patient communication, documentation, and discharge summary preparation.

RESEARCH EXPERIENCES

AI and LLMs in Health and Social Science | Washington University in St. Louis

2022 – 2026

Research Assistant

- Supported the development and applied use of Empathy Helper, an LLM-enabled simulated-client chatbot used to train social work students (Advisor: Ryan Lindsay).
- Developed end-to-end workflows for data curation, prompting, RAG, and evaluation; supported manuscript preparation and presentations (Advisor: Ruopeng An).
- Built transformer-based models to estimate momentary affect intensity from smartphone-based passive sensing data (Advisor: Renee Thompson).

Clonal Hematopoiesis and Risk of Incident Lung Cancer | Washington University School of Medicine

2022

Research Assistant, Division of Public Health Sciences

- Prospectively examined the association between clonal hematopoiesis and subsequent risk of lung cancer. (Yin Cao's Lab)

The Role of GSDME in Breast Cancer Multidrug Resistance | Fudan University

2017 – 2020

Undergraduate Project Investigator, Department of Pathology

- Developed doxorubicin, carboplatin, and 5-FU resistant MCF-7 and MDA-MB-231 breast cancer cell lines; quantified GSDME expression and pyroptosis via qPCR and Western blot.

The Role and Mechanism of RacGAP1 Regulated Mitochondrial Quality Control in Breast Cancer Invasion and Metastasis | Fudan University

2017 – 2018

Research Assistant, Department of Pathology

- Supported breast cancer invasion and metastasis research through literature review and core wet-lab assays including cell culture, lentiviral packaging, Western blot, PCR, gel electrophoresis, and RNA/DNA/protein extraction.

LEADERSHIP & SERVICE

- Graduate Student Affairs Advisory Board, Member (2023–Present)
- Co-organizer, AI Interest Group (2023–2024). Led biweekly research-sharing sessions on AI applications and advances.
- Peer Mentor, Brown School Peer2Peer Program (2022–2023)